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登陆台风“彩虹”（2015）近岸海洋响应及其对快速增强海洋下垫面条件的指示

Near-shore oceanic response to landfalling Typhoon Mujigae (2015) and its implications for oceanic substrate conditions during rapid intensification

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摘要: 热带气旋（TC）强度变化预报中的海气相互作用过程长期存在不确定性。2015 年台风“彩虹”在南海北部登陆前经历了快速增强（RI），已有研究主要从大气动力学角度揭示了其动能/势能转换、内核结构调整和边界层恢复等机制，但近岸海洋下垫面在该阶段所扮演的角色尚不明确。本文以“彩虹”为例，利用 CMA-STI 最佳路径、ERA5 逐小时 10 m 风场、MUR 高分辨率海表温度（SST）以及 GLORYS12 海洋再分析资料，围绕“风场配置---SST 响应---次表层结构---Ekman 输送”四维证据链，诊断其登陆前近岸海洋响应的性质。结果表明：登陆前较长时间内，近岸诊断区出现持续的下涌有利沿岸风，风速达到较强水平，并在登陆前数小时达到峰值；相应的风应力和向岸 Ekman 输送峰值均达较高强度，正向累计输送指数显著。MUR SST 显示，近岸区靠岸过程中平均增暖，增暖格点占绝大多数，至登陆日仅较靠岸前出现极弱冷却，表现举与外海冷却形成空间对照的海温维持特征。次表层多个代表性等温线在近岸区均呈下压趋势，为下涌型垂向结构调整提供辅助证据。

综合四维证据判断，近岸下涌型响应使 SST 未出现典型台风冷尾流式的大幅冷却，有利于维持海气焓通量供应，从而为“彩虹”登陆前强度维持提供了边际有利的海洋下垫面条件。本文结论限定为基于再分析与遥感资料的诊断性推断，不等同于对 RI 因果机制的严格验证。

关键词: 台风彩虹；沿海下涌；海表温度维持；快速增强；南海北部

Abstract: The air-sea interaction process in tropical cyclone (TC) intensity prediction has long been uncertain. Typhoon Mujigae (2015) underwent rapid intensification (RI) before making landfall in the northern South China Sea. This study uses CMA-STI best track data, ERA5 hourly 10 m wind fields, MUR high-resolution sea surface temperature (SST), and GLORYS12 ocean reanalysis data to diagnose the nature of near-shore ocean response before landfall. Results show persistent downwelling-favorable alongshore winds, significant Ekman transport, near-shore SST maintenance, and subsurface isotherm deepening. Comprehensive evidence indicates the near-shore downwelling-type response prevented SST from experiencing significant cooling, providing marginally favorable ocean substrate conditions for Mujigae pre-landfall intensity maintenance.

Keywords: Typhoon Mujigae; coastal downwelling; SST maintenance; rapid intensification; northern South China Sea

1 引言

热带气旋（TC）的强度变化受海表温度、上层海洋热含量、边界层湍流交换、环境垂直风切变及内核动力热力过程的共同调控[1-13]。其中，海气焓通量（包括感热通量与潜热通量）为 TC 的维持和增强提供根本能量来源，而台风自身强迫引起的上层海洋垂向混合、近惯性内波响应及冷尾流过程则可通过降低海表温度（SST）形成负反馈，抑制海气焓通量供应并削弱 TC 强度[1-4]。因此，海洋既是 TC 的能量库，也是近岸强度变化预报中最难被准确刻画的下垫面边界条件[5-6]。在全球变暖背景下，TC 快速增强事件的频率与强度变化已成为海洋气象学的前沿议题，海洋下垫面热力条件的演变在其中发挥着关键作用[7-13]。

陆架海域的台风—海洋相互作用较开阔大洋表现出更强的地域性特征[14-21]。一方面，在宽陆架和强层结环境中，台风前侧浅水区的强剪切混合可导致剧烈海表冷却（幅度可达数度至十余度），显著削弱海气热量供应并引起登陆前 TC 强度衰减[14-16]。另一方面，Gramer 等[18]提出了“沿海下涌增强登陆飓风”的机制：当 TC 靠近海岸时，其右前象限的近岸风场驱动向岸 Ekman 输送，将离岸暖水向近岸堆积并通过下涌维持陆架暖 SST，从而削弱甚至逆转冷却效应，增强海气焓通量并使飓风在登陆前仍能维持或增强。该机制

为大西洋和墨西哥湾较宽陆架环境下飓风登陆前的异常增强提供了有力解释。然而，向南海北部窄陆架外推时，陆架宽度不足可能制约完整横向环流的建立时间和调整空间[19-21]，因此宽陆架下涌机制在窄陆架中的适用性仍是一个开放问题。

南海北部粤西—雷州半岛东侧海域具有岸线弯曲、陆架狭窄的地形特征——50、100 和 200 m 等深线距岸较近，局地水深梯度与岬角地形约束显著。该区域既受南海季风、黑潮入侵及中尺度涡等大尺度背景影响[25-31]，又频繁受到登陆台风的直接影响。近年来，已有研究发现不同海盆间 TC 增强的海洋预条件存在显著跨海盆差异，陆架宽度和层结状态是关键控制因子[42-44]。对于窄陆架而言，内陆架风驱升降流响应的理论分析表明，地形坡度将有效约束 Ekman 调整的时间与空间尺度[19]，因而不能简单将宽陆架机制套用于南海北部窄陆架环境。

2015 年第 22 号台风“彩虹”（Mujigae）在南海北部向西北移动并逼近雷州半岛过程中经历了显著增强，由台风级增强至强台风级乃至超强台风级。已有研究从异常暖 SST 背景、内核对流组织、能量转换（动能/势能收支）以及边界层过程等角度对其 RI 进行了系统分析[22-24, 39-40]。其中，Zhang 等[25]的动能/势能收支分析表明，“彩虹”的 RI 主要由大气内部的能量转换过程驱动。然而，上述研究几乎完全聚焦于大气动力学和热力学框架，近岸海洋下垫面——特别是登陆前最后阶段近岸 SST 的状态及其对海气焔通量供应的维持能力——至今仍未被纳入“彩虹”RI 的讨论范畴。

由此产生一个核心科学问题：在“彩虹”登陆前，粤西—雷州半岛东侧窄陆架海域是否出现了沿海下涌型海洋响应？若存在，这种响应是否使近岸 SST 避免了典型冷尾流式的大幅冷却，进而为登陆前强度维持提供了边际有利的海洋下垫面条件？对这一问题的回答，不仅有助于补充现有 RI 多因子框架中所缺失的“近岸海洋边界条件”环节，也可为评估 Gramer[18]机制在窄陆架环境中的适用性提供新的诊断案例。

基于上述问题，本文以台风“彩虹”为例，利用多源再分析与遥感资料，围绕“风场配置—海表温度响应—一次表层结构—Ekman 输送”建立四维现象证据链，检验南海北部窄陆架环境下是否存在与沿海下涌相兼容的可辨识信号，并在此基础上推断其对“彩虹”登陆前 RI 的海洋下垫面指示意义。本文定位为诊断性个例研究，不将结果表述为严格的动力学验证；任何关于“下涌→SST 维持→强度变化”的因果推断，均有赖于后续高分辨率海气耦合敏感性试验的检验。

2 数据与方法

3 结果分析

4 讨论

5 结论

本文利用 CMA-STI 最佳路径、ERA5 逐小时风场、MUR SST 和 GLORYS12 海洋再分析资料,以 2015 年台风“彩虹”为例,围绕“风场配置—SST 响应—一次表层结构—Ekman 输送”四维证据链,诊断了南海北部粤西—雷州半岛东侧窄陆架海域的登陆前近岸海洋响应特征。主要结论如下:

(1) 台风“彩虹”登陆前,粤西—雷州半岛东侧近岸诊断区出现持续的下涌有利沿岸风。登陆前较长时段内 U_{dw} 持续为正且强度较强,登陆前数小时达到峰值,为向岸 Ekman 输送和近岸暖水堆积提供了必要的动力条件。

(2) MUR SST 显示近岸诊断区在靠岸过程中表现为平均增暖,绝大多数格点呈增暖趋势;至登陆日仅较靠岸前出现极弱冷却。近岸区与 100--200 m 等深线外侧深水冷却区形成“近岸维持、外海冷却”的显著空间对照,表明近岸 SST 未经历典型台风冷尾流式的大幅降温。

(3) GLORYS12 次表层温度剖面显示,近岸段多个代表性等温线均呈一致下压趋势,方向一致(均为正值),为下涌型垂向结构调整提供了辅助性支持证据。

(4) 下涌有利风应力 τ_{dw} 和向岸 Ekman 输送 M_E 均达到较强水平,正向累计输送指数 I_E 持续累积增长,其累积增长阶段与近岸 SST 弱增暖/维持阶段在时间上高度对应。

(5) 综合四维证据判断,台风“彩虹”在南海北部窄陆架环境下登陆前存在可辨识的、但强度有限的沿海下涌型海洋响应。该响应通过维持近岸 SST、避免大幅冷却,有利于维持海气焓通量供应,从而为“彩虹”登陆前强度维持提供了边际有利的海洋下垫面条件。需要明确指出的是,本结论定位为基于再分析与遥感资料的诊断性推断,不等同于对 RI 因果机制的严格验证。对上述推论的确证,尚有赖于高分辨率海气耦合敏感性试验和多台风统计对比研究。

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资料	时空分辨率	主要变量	用途	来源
CMA-STI 最佳路径	6 h	台风中心位置、强度	确定路径等级和参考登陆时刻	[35-36]
ERA5 再分析	逐小时，约 0.25°	10 m 风场	近岸风场、沿岸风、风应力和 Ekman 输送诊断	[32]
MUR SST L4	日尺度，约 0.01°	海表温度	靠岸前后近岸 SST 空间差异	[33]
GLORYS12	日尺度，约 1/12°，50 层	温度剖面	次表层温度结构和等温线深度	[34]
水深资斐	公开全球水深网格	地形、水深	提取 50、100、200 m 等深线	---

表 1 本文使用的主要资料及用途

诊断量	数值或范围	物理含义
U_dw 正值持续时间	10-03 06 时--10-04 03 时 UTC, 约 22 h	近岸风场长期处于下涌有利几何配置
U_dw 均值/峰值	9.91 / 13.04 m s ⁻¹	强沿岸风可产生显著向岸 Ekman 输送
τ_dw 峰值	0.525 N m ⁻²	下涌有利风应力在登陆前 9 h 最强
M_E 峰值	9.84 m ² s ⁻¹	单位岸线宽度向岸输送较强
I_E 正向累计输送指数	4.42×10 ⁵ m ²	下涌有利过程具有持续累积效应
近岸 ΔSST	+0.059 °C (10-03 减 10-02) ; -0.049 °C	靠岸过程中近岸海温维持或弱增暖

表 2 近岸诊断区主要风场、Ekman 输送与 SST 诊断结果

参考文献